Fourier sine series

Now that we've looked at the Fourier cosine series, the Fourier sine series will just be an odd extension of all those same ideas.

We just need to start with the understanding that, while the cosine function is even (it's symmetric about the y-axis and satisfies $f(-x) = f(x)$), the sine function is odd (it's symmetric about the origin and satisfies $f(-x) = -f(x)$).

The sine series

If we try to find the Fourier series representation,

$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\ dx$$

$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\ dx \qquad n = 1,\ 2,\ 3,\ \ldots$$

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\ dx \qquad n = 1,\ 2,\ 3,\ \ldots$$

of an odd function on $-L \leq x \leq L$, we'll discover that the coefficients A_0 and A_n are 0, such that the Fourier series of the odd function will simplify to just

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$$f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx \qquad\qquad n = 1, 2, 3, \ldots$$

Because this is a Fourier series in terms of only sine functions, we call it a **Fourier sine series**. What this means is that, when we want to find the Fourier series representation of an odd function, we actually only need to find its Fourier sine series. If we know the function is odd, we don't have to bother computing A_0 or A_n or trying to find the cosine parts of the Fourier series representation.

Let's do an example with an odd function to prove that this is true.

Example

Find the Fourier series and Fourier sine series of $f(x) = x$ on $-L \leq x \leq L$, and show that they are equivalent.

Let's use the original definition of a Fourier series, which means we'll start by calculating A_0, A_n, and B_n. For A_0 we get

$$A_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\,dx$$

$$A_0 = \frac{1}{2L}\int_{-L}^{L} x\,dx$$

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This is the integral of an odd function on a symmetric interval, which means the value of the integral must be 0.

$$A_0 = \frac{1}{2L}(0)$$

$$A_0 = 0$$

For A_n we get

$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx$$

$$A_n = \frac{1}{L}\int_{-L}^{L} x\cos\left(\frac{n\pi x}{L}\right)\,dx$$

The function $f(x) = x$ is odd, and the cosine function is even, and the product of an odd function and an even function is itself an odd function. Which means this is the integral of an odd function on a symmetric interval, which means the value of the integral must be 0.

$$A_n = \frac{1}{L}(0)$$

$$A_n = 0$$

And for B_n we get

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{1}{L}\int_{-L}^{L} x\sin\left(\frac{n\pi x}{L}\right)\,dx$$

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Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\,dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = \frac{1}{L}\left[-x\left(\frac{L}{n\pi}\right)\cos\left(\frac{n\pi x}{L}\right) + \int\left(\frac{L}{n\pi}\right)\cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_{-L}^{L}$$

$$B_n = \frac{L}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right) - \frac{x}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Bigg|_{-L}^{L}$$

$$B_n = \frac{L}{(n\pi)^2}\sin\left(\frac{n\pi L}{L}\right) - \frac{L}{n\pi}\cos\left(\frac{n\pi L}{L}\right)$$

$$-\left[\frac{L}{(n\pi)^2}\sin\left(\frac{-n\pi L}{L}\right) + \frac{L}{n\pi}\cos\left(\frac{-n\pi L}{L}\right)\right]$$

$$B_n = \frac{L}{(n\pi)^2}\sin(n\pi) - \frac{L}{n\pi}\cos(n\pi) - \left[\frac{L}{(n\pi)^2}\sin(-n\pi) + \frac{L}{n\pi}\cos(-n\pi)\right]$$

$$B_n = \frac{L}{(n\pi)^2}\sin(n\pi) - \frac{L}{n\pi}\cos(n\pi) + \frac{L}{(n\pi)^2}\sin(n\pi) - \frac{L}{n\pi}\cos(n\pi)$$

For $n = 1, 2, 3, \ldots$, $\sin(n\pi) = 0$ and $\cos(n\pi) = (-1)^n$, so the expression for B_n simplifies to

$$B_n = -\frac{2L}{n\pi}(-1)^n$$

$$B_n = \frac{(-1)^{n+1}2L}{n\pi}$$

Then the Fourier series representation of $f(x) = x$ on $-L \le x \le L$ is

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$$f(x) = A_0 + \sum_{n=1}^{\infty} A_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = 0 + \sum_{n=1}^{\infty} (0)\cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}2L}{n\pi} \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{2L}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin\left(\frac{n\pi x}{L}\right)$$

Now let's use the Fourier sine series definition, which means we'll start by calculating B_n.

$$B_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) \, dx$$

$$B_n = \frac{2}{L} \int_0^L x \sin\left(\frac{n\pi x}{L}\right) \, dx$$

Use integration by parts with $u = x$, $du = dx$, $dv = \sin(n\pi x/L)\, dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = \frac{2}{L} \left[-x \left(\frac{L}{n\pi}\right) \cos\left(\frac{n\pi x}{L}\right) + \int \left(\frac{L}{n\pi}\right) \cos\left(\frac{n\pi x}{L}\right) \, dx \right] \Bigg|_0^L$$

$$B_n = \frac{2L}{(n\pi)^2} \sin\left(\frac{n\pi x}{L}\right) - \frac{2x}{n\pi} \cos\left(\frac{n\pi x}{L}\right) \Bigg|_0^L$$

$$B_n = \frac{2L}{(n\pi)^2} \sin\left(\frac{n\pi L}{L}\right) - \frac{2L}{n\pi} \cos\left(\frac{n\pi L}{L}\right)$$

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$$-\left[\frac{2L}{(n\pi)^2}\sin\left(\frac{n\pi(0)}{L}\right)-\frac{2(0)}{n\pi}\cos\left(\frac{n\pi(0)}{L}\right)\right]$$

$$B_n=\frac{2L}{(n\pi)^2}\sin(n\pi)-\frac{2L}{n\pi}\cos(n\pi)$$

For $n=1, 2, 3, \ldots$, $\sin(n\pi)=0$ and $\cos(n\pi)=(-1)^n$, so the expression for B_n simplifies to

$$B_n=-\frac{2L}{n\pi}(-1)^n$$

$$B_n=\frac{(-1)^{n+1}2L}{n\pi}$$

Then the Fourier sine series of $f(x)=x$ on $-L\le x\le L$ is

$$f(x)=\sum_{n=1}^{\infty}B_n\sin\left(\frac{n\pi x}{L}\right)$$

$$f(x)=\sum_{n=1}^{\infty}\frac{(-1)^{n+1}2L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)$$

$$f(x)=\frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin\left(\frac{n\pi x}{L}\right)$$

Because $f(x)=x$ is an odd function on $-L\le x\le L$, we get equivalent representations of $f(x)$ for both its Fourier series and its Fourier sine series.

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Functions that aren't odd

While it's easy to model odd functions with a Fourier sine series, we can also use the sine series to model functions that aren't odd.

We start with the idea that, for any function defined on $0 \leq x \leq L$, we can find its odd extension on $-L \leq x \leq 0$. We talked about the even and odd extensions of a function when we looked at periodic extensions earlier. But as a reminder, given $f(x)$ defined on $0 \leq x \leq L$, we create its reflection about the origin on $-L \leq x \leq 0$, thereby creating a new function $g(x)$,

$$g(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ -f(-x) & -L \leq x < 0 \end{cases}$$

which is itself an odd function defined on $-L \leq x \leq L$.

Because $g(x)$ is odd, we know we can find the Fourier sine series representation of $g(x)$ (or its Fourier series representation, in general) on $-L \leq x \leq L$.

But then, because $g(x)$ and $f(x)$ are equivalent on $0 \leq x \leq L$, we know that the Fourier sine series of $g(x)$ (or the Fourier series of $g(x)$), will be equivalent to the Fourier sine series of $f(x)$ (or the Fourier series of $f(x)$), as long as we restrict the interval of definition to $0 \leq x \leq L$, instead of $-L \leq x \leq L$.

So let's do another example, this time where we find the Fourier sine series of a function that isn't odd.

Example

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Find the Fourier sine series of $f(x) = x^2$ on $0 \leq x \leq L$.

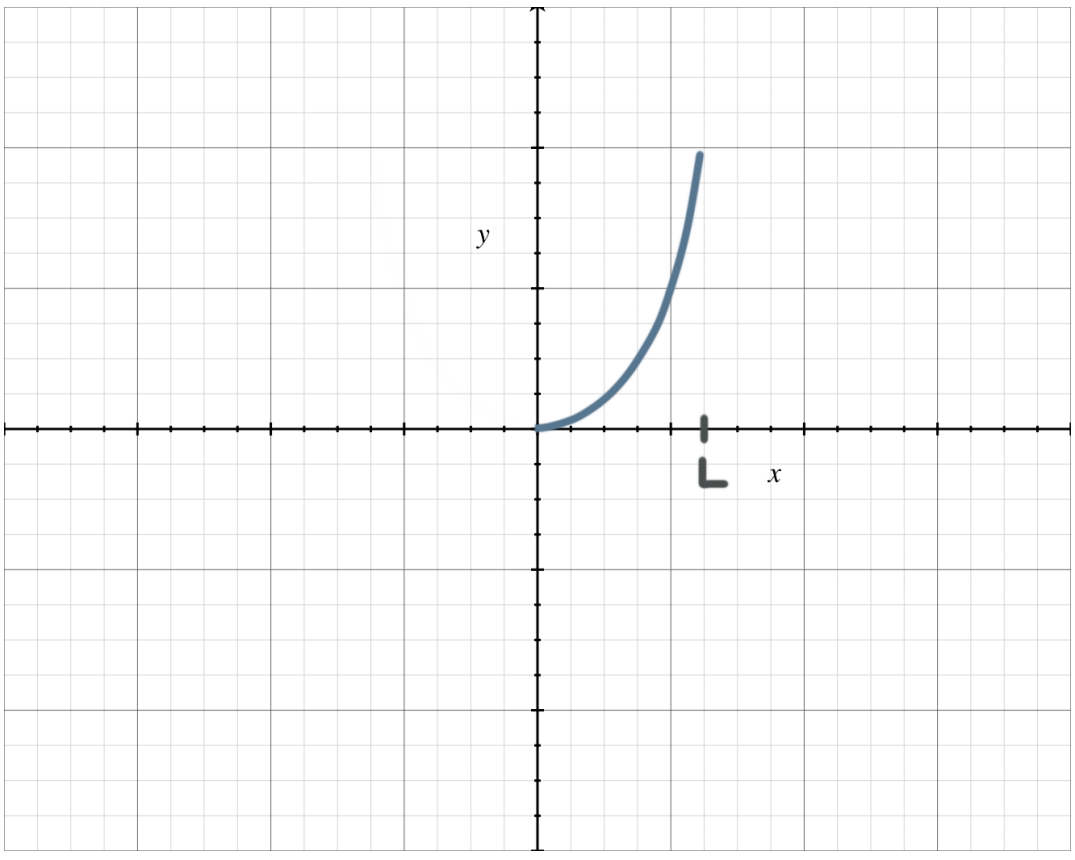
This function is even, so we'll start by finding the odd extension of the even function.

$$g(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ -f(-x) & -L \leq x < 0 \end{cases}$$

$$g(x) = \begin{cases} x^2 & 0 \leq x \leq L \\ -((-x)^2) & -L \leq x < 0 \end{cases}$$

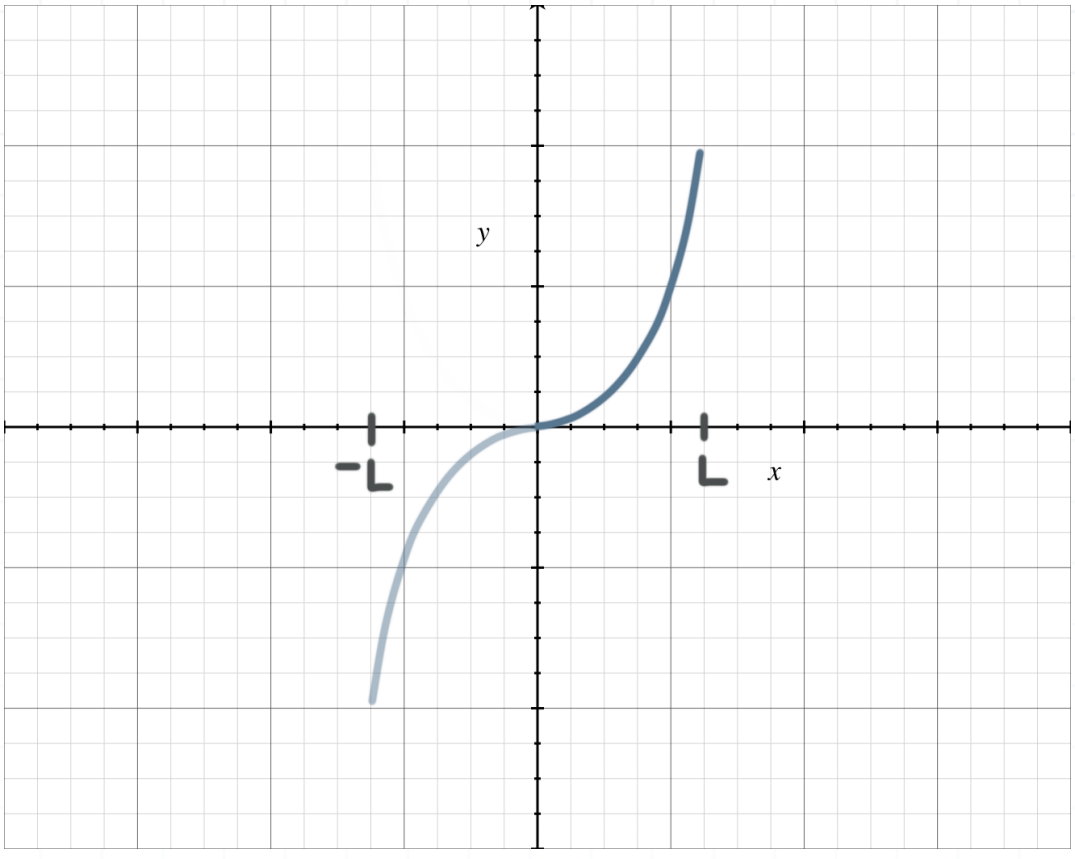
$$g(x) = \begin{cases} x^2 & 0 \leq x \leq L \\ -x^2 & -L \leq x < 0 \end{cases}$$

A sketch of the even function $f(x)$ on $0 \leq x \leq L$ is



and the sketch of its odd extension $g(x)$ on $-L \leq x \leq L$ is

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Because the odd extension is an odd function, we can find its Fourier sine series representation, starting with calculating B_n.

$$B_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx$$

$$B_n = \frac{2}{L}\int_0^L x^2\sin\left(\frac{n\pi x}{L}\right)\,dx$$

Use integration by parts with $u = x^2$, $du = 2x\ dx$, $dv = \sin(n\pi x/L)\ dx$, and $v = -(L/n\pi)\cos(n\pi x/L)$.

$$B_n = \frac{2}{L}\left[-x^2\left(\frac{L}{n\pi}\right)\cos\left(\frac{n\pi x}{L}\right) + \int \frac{2Lx}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

$$B_n = \frac{2}{L}\left[-\frac{Lx^2}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{2L}{n\pi}\int x\cos\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

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Use integration by parts with $u = x$, $du = dx$, $dv = \cos(n\pi x/L)\,dx$, and $v = (L/n\pi)\sin(n\pi x/L)$.

$$B_n = \frac{2}{L}\left[-\frac{Lx^2}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\right.$$

$$\left.+\frac{2L}{n\pi}\left[\frac{Lx}{n\pi}\sin\left(\frac{n\pi x}{L}\right) - \int \frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\,dx\right]\right]\Bigg|_0^L$$

$$B_n = \frac{2}{L}\left[-\frac{Lx^2}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{2L^2x}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right) - \frac{2L^2}{(n\pi)^2}\int \sin\left(\frac{n\pi x}{L}\right)\,dx\right]\Bigg|_0^L$$

$$B_n = \frac{2}{L}\left[-\frac{Lx^2}{n\pi}\cos\left(\frac{n\pi x}{L}\right) + \frac{2L^2x}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right) + \frac{2L^3}{(n\pi)^3}\cos\left(\frac{n\pi x}{L}\right)\right]\Bigg|_0^L$$

$$B_n = \frac{4Lx}{(n\pi)^2}\sin\left(\frac{n\pi x}{L}\right) + \frac{4L^2}{(n\pi)^3}\cos\left(\frac{n\pi x}{L}\right) - \frac{2x^2}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\Bigg|_0^L$$

Evaluate over the interval.

$$B_n = \frac{4L^2}{(n\pi)^2}\sin\left(\frac{n\pi L}{L}\right) + \frac{4L^2}{(n\pi)^3}\cos\left(\frac{n\pi L}{L}\right)$$

$$-\frac{2L^2}{n\pi}\cos\left(\frac{n\pi L}{L}\right) - \frac{4L^2}{(n\pi)^3}\cos\left(\frac{n\pi(0)}{L}\right)$$

$$B_n = \frac{4L^2}{(n\pi)^2}\sin(n\pi) + \frac{4L^2}{(n\pi)^3}\cos(n\pi) - \frac{2L^2}{n\pi}\cos(n\pi) - \frac{4L^2}{(n\pi)^3}$$

For $n = 1, 2, 3, \ldots$, $\sin(n\pi) = 0$ and $\cos(n\pi) = (-1)^n$, so the expression for B_n simplifies to

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$$B_n = \frac{4L^2}{(n\pi)^3}(-1)^n - \frac{2L^2}{n\pi}(-1)^n - \frac{4L^2}{(n\pi)^3}$$

$$B_n = \frac{(-1)^n 4L^2 + (-1)^{n+1} 2L^2 (n\pi)^2 - 4L^2}{(n\pi)^3}$$

Then the Fourier sine series of $f(x) = x^2$ on $0 \le x \le L$ is

$$f(x) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \sum_{n=1}^{\infty} \frac{(-1)^n 4L^2 + (-1)^{n+1} 2L^2 (n\pi)^2 - 4L^2}{(n\pi)^3} \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \frac{2L^2}{\pi^3} \sum_{n=1}^{\infty} \frac{2(-1)^n + (-1)^{n+1} (n\pi)^2 - 2}{n^3} \sin\left(\frac{n\pi x}{L}\right)$$

Even though the function $f(x) = x^2$ isn't odd, we were still able to find a Fourier sine series to represent it on $0 \le x \le L$.

Convergence of the sine series

We know from earlier that a Fourier series will converge to the function $f(x)$ that it represents as long as $f(x)$ is piecewise smooth and its periodic extension is continuous.

Otherwise the Fourier series will converge to the average of the one-sided limits if the periodic extension has a jump discontinuity between periods.

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But now we want to specifically address the convergence of a Fourier sine series.

If we're looking at the Fourier sine series of an odd function on $-L \leq x \leq L$, then it'll converge to $f(x)$ just like a normal Fourier series. Of course, that's because the Fourier sine series of an odd function is just the special case of the general Fourier series, where $A_0 = 0$ and $A_n = 0$ and the cosine series parts of the Fourier series disappear.

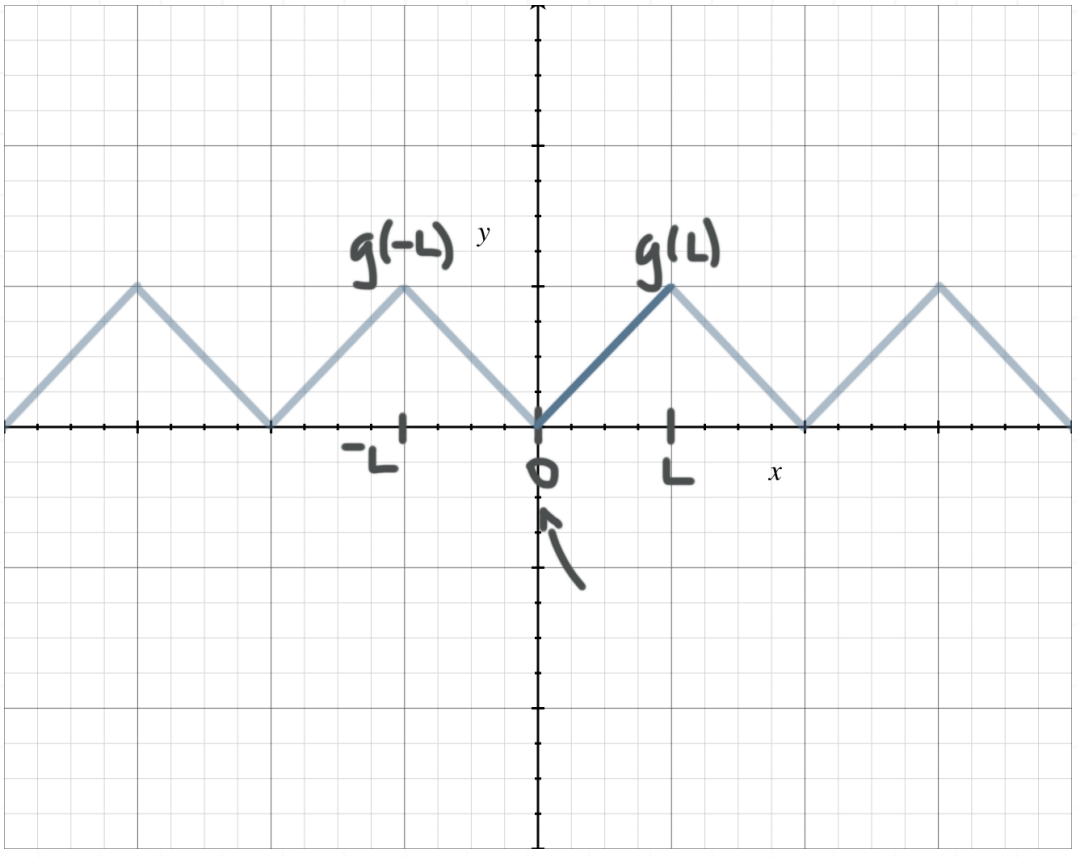
But what about the convergence of the Fourier sine series of a function that isn't odd?

Well remember, if we take a function defined on $0 \leq x \leq L$ and build its odd extension, that odd extension is now an odd function $g(x)$ on $-L \leq x \leq L$, and we can therefore find the Fourier sine series of that odd $g(x)$ function on $-L \leq x \leq L$.

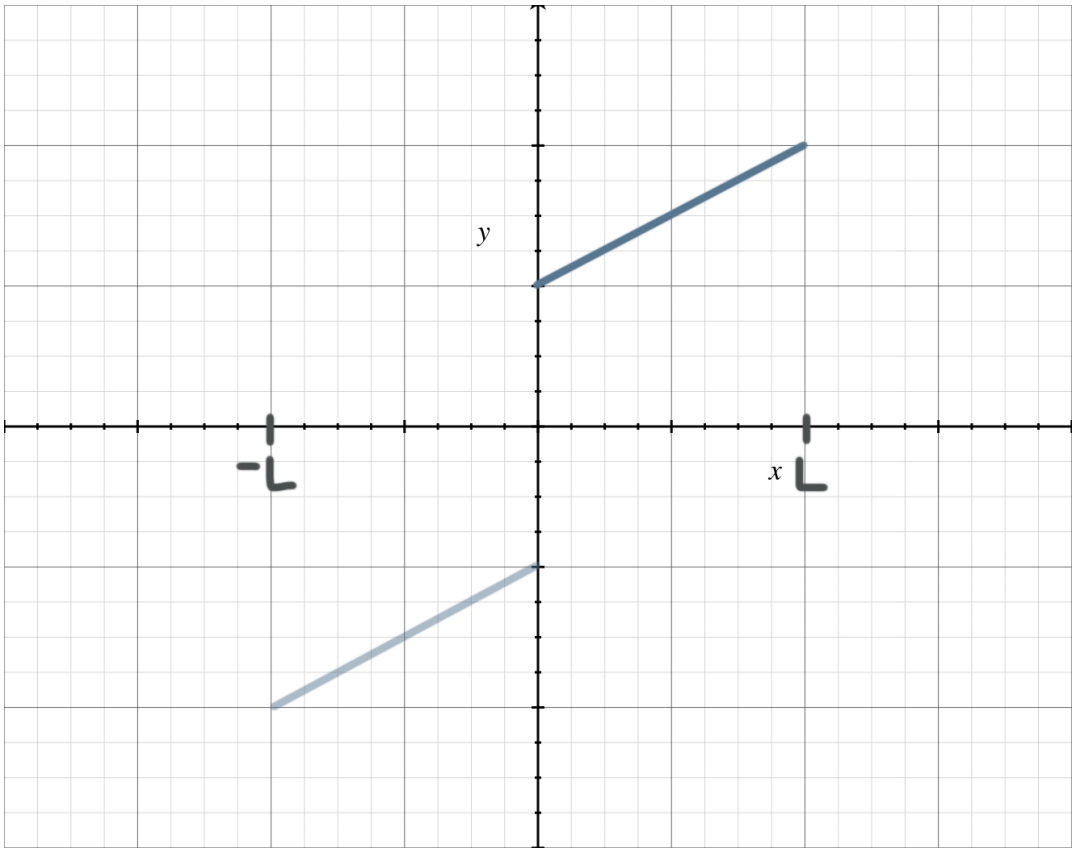
We saw how even extensions were always continuous (assuming no discontinuity within the interval $0 \leq x \leq L$), because their symmetry means that the two one-sided limits at $x = 0$ are always equivalent, and that $g(-L) = g(L)$, which means the one-sided limits between periods are equivalent.

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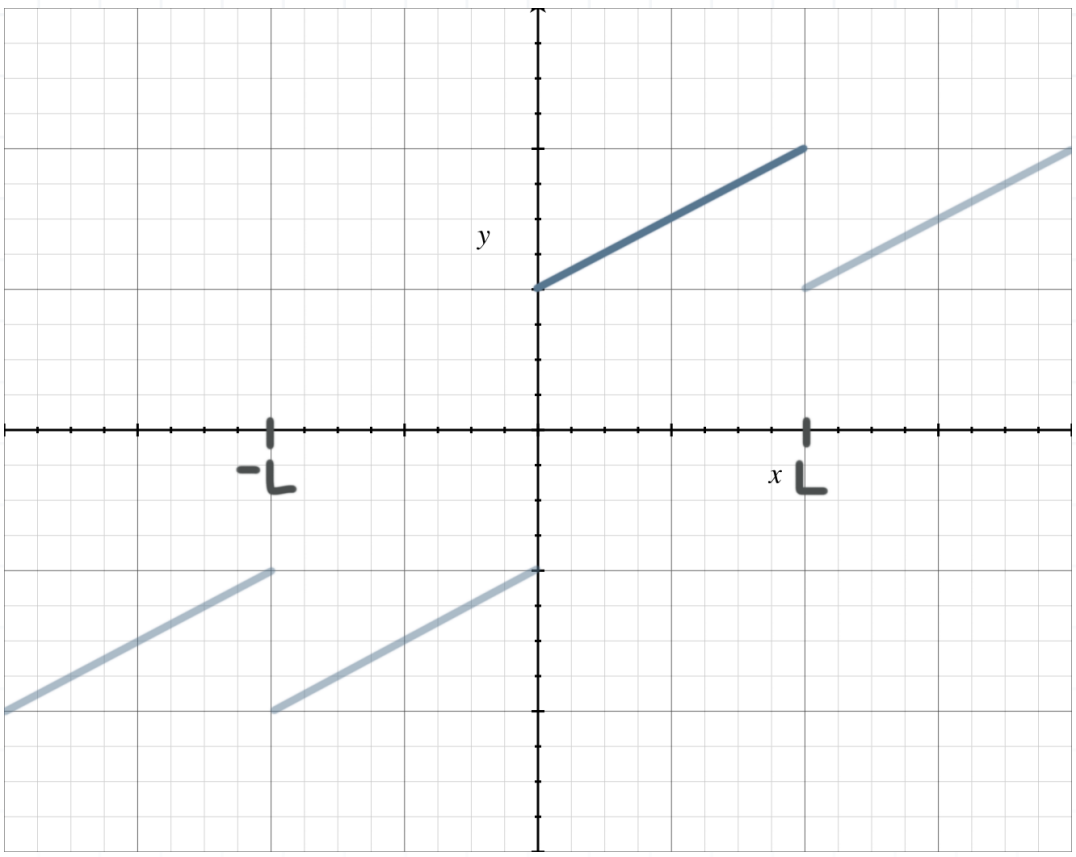


But this isn't necessarily the case with odd functions. For instance, here's the sketch of an odd extension with a jump discontinuity at $x = 0$,



and here's the sketch of a periodic extension of with a jump discontinuity between each period.

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In fact, the only way the periodic extension of the odd function will be continuous is when both $f(0) = 0$ and $f(L) = 0$. If these two requirements are met, then the periodic extension is continuous.

Which means, based on what we already know about the convergence of a Fourier series, that the Fourier series representation of $g(x)$ will converge to $g(x)$ on $-L \leq x \leq L$. The Fourier sine series, as a special case of the Fourier series, will also converge to $g(x)$ on $-L \leq x \leq L$.

And because the odd extension $g(x)$ is equivalent to the original function $f(x)$ on $0 \leq x \leq L$, we can conclude that the Fourier sine series representation of $f(x)$ will converge to $f(x)$ on $0 \leq x \leq L$.

Therefore,

- the Fourier sine series of an odd function $f(x)$ will converge to $f(x)$ on $-L \leq x \leq L$.

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- the Fourier sine series of a function $f(x)$ that isn't odd will converge to $f(x)$ on $0 \leq x \leq L$, provided $f(x)$ is continuous on $0 \leq x \leq L$, and that $f(0) = 0$ and $f(L) = 0$.

The only exception is if the function $f(x)$ contains a jump discontinuity somewhere within $0 \leq x \leq L$. But just like with the general Fourier series, the Fourier sine series will converge to the average of the one-sided limits at that jump discontinuity.

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